



QGIS PROCESSING PLUGIN

DWG / DXF Converter

Reliable, bidirectional CAD file conversion for QGIS, built for automated pipelines.

What This Plugin Does

DWG / DXF Converter adds AutoCAD DWG ↔ DXF conversion to QGIS. It does this by wrapping the free ODA File Converter tool inside four QGIS Processing algorithms, so conversion becomes a normal part of a QGIS workflow instead of a manual detour through a separate application.

Because it is built entirely on the Processing framework rather than a custom dialog, it works everywhere Processing works: interactively from the toolbox, in batch over many files at once, chained inside Processing models, scripted with `processing.run()`, or run completely headlessly from the command line with `qgis, process`, making it equally suited to a one-off conversion or an unattended nightly pipeline.

The Problem It Solves

DWG is a closed, proprietary Autodesk format. QGIS and the underlying GDAL/OGR libraries cannot read or write it natively, there is no open-source path around this. Anyone working with DWG survey plans, architectural drawings, or utility networks inside QGIS has traditionally had to convert files by hand in a separate desktop application before every import, and convert them back by hand before handing work off to AutoCAD users.

This plugin automates that conversion in both directions and folds it directly into the QGIS/Processing ecosystem, removing the manual round-trip.

Key Features



Bidirectional conversion

Convert DWG → DXF for use in QGIS, and DXF → DWG to hand work back to AutoCAD users , four algorithms cover both directions, singly or in bulk.



Bulk / incremental conversion

Convert an entire folder in one pass. Files already converted and unchanged are skipped automatically, so re-running the same folder repeatedly is fast and safe.



Automation-first design

No custom dialog to click through. Every algorithm runs from `qgis_process`, PyQGIS scripts, Processing models, or QGIS's built-in batch runner.



Built-in reliability safeguards

A configurable timeout kills a hung process instead of letting a pipeline stall; corrupt or empty outputs are detected and reported rather than silently accepted.



Broad compatibility

Works across QGIS 3.x and 4.x, and across Windows, macOS, and Linux, including headless Linux servers via automatic xvfb support.

How It Works

The plugin registers a QGIS Processing provider containing four algorithms, all built on two shared, well-tested base classes — one for single-file conversion, one for folder/bulk conversion. Each algorithm assembles a call to the ODA File Converter command-line tool, runs it under Qt's QProcess, streams its output into the Processing log, and validates the result before handing control back to QGIS.

This shared-base design means every algorithm, regardless of direction or single/bulk mode, gets the same reliability behaviour: timeout protection, corrupt-output detection, and clear error messages, with no duplicated logic to fall out of sync.

Requirements

- QGIS 3.0 or later (including QGIS 4.x)
- ODA File Converter, free download from opendesign.com; installed separately, as its license does not permit redistribution
- On headless Linux only: xvfb, since ODA File Converter needs a display even in batch mode

Credits



[Developed by Kelvin Moseti](#)